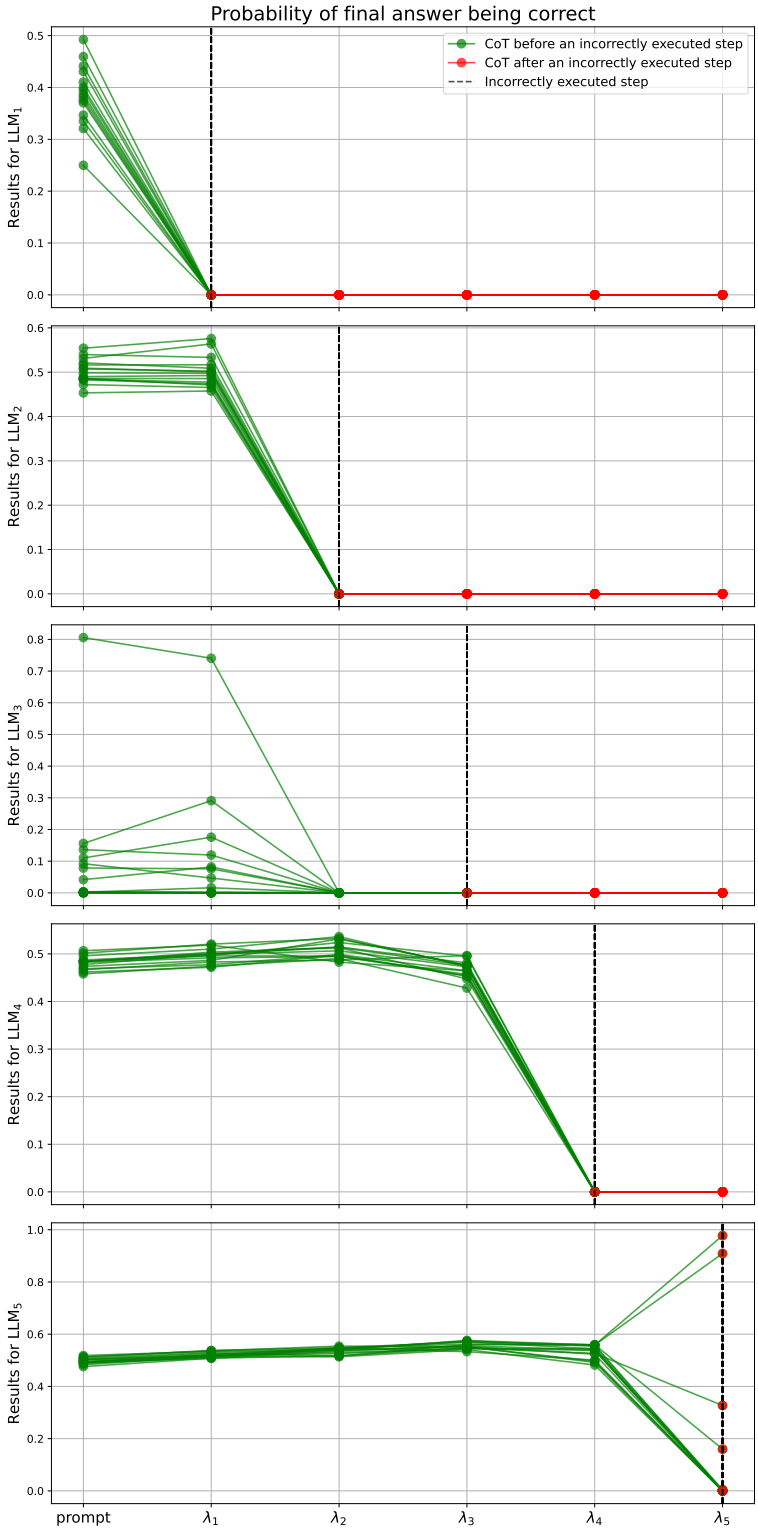

Figure 6 Toy data results: ORM's probability of correctness after each step for 15 randomly chosen prompts with wrong final answers.

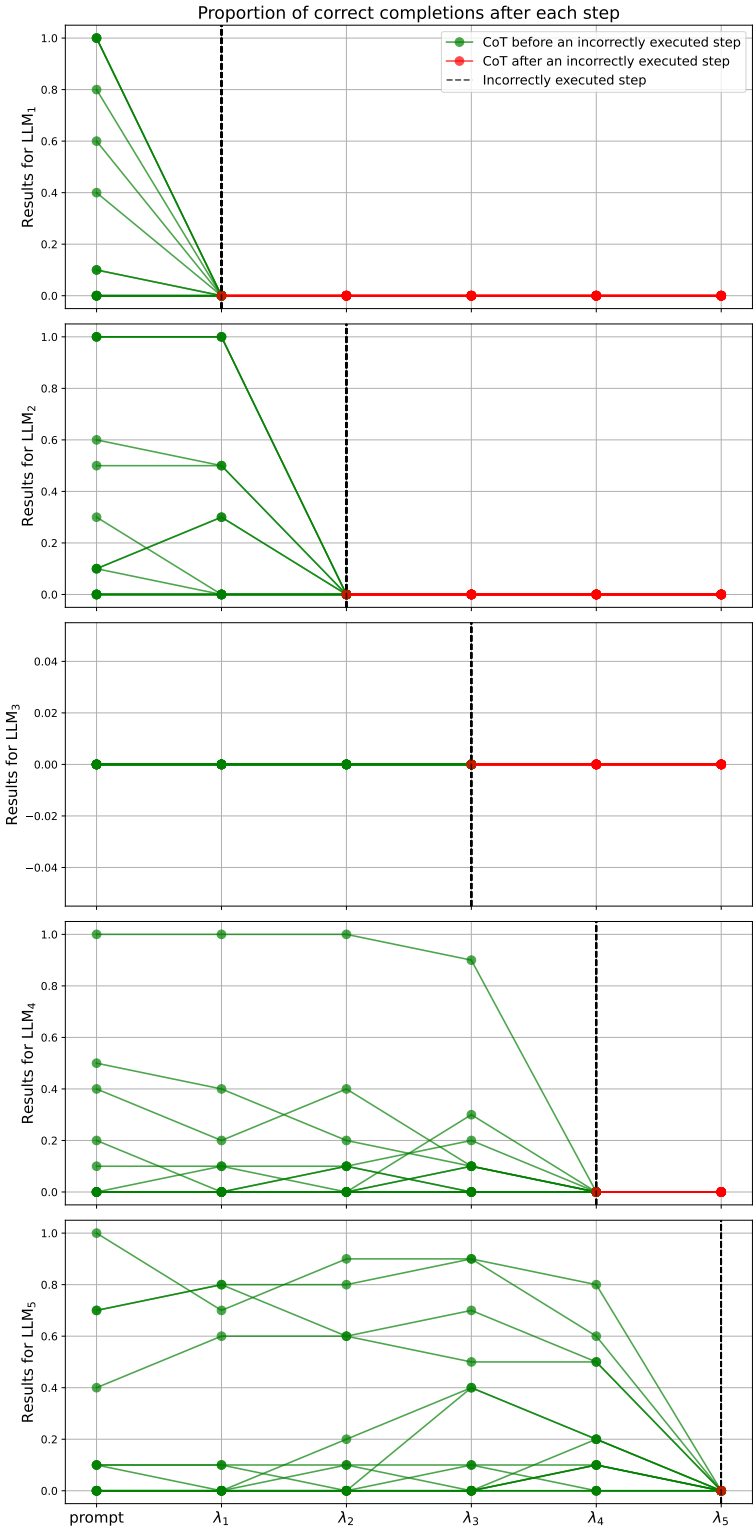

Figure 7 Toy data results: Math-Shepherd's proportion of correct completions from each step for 15 randomly chosen prompts with wrong final answers.

C.2.1 Training Data for the Supervisor Model

The *supervisor model* plays a crucial role in evaluating the intermediate steps in the Chain-of-Thought (CoT) reasoning. The model is designed to approximate the probability of arriving at the correct final result after any given step in the CoT process. To train this model, we fine-tune it using a dataset composed of generated CoT steps concatenated with the correct final result.

Model Generation Example: Consider the following example of a model-generated response:

```
x = 51290.0, y = 90718.0. Please calculate the following:
1. 3x
2. 2y
3. 3x + 2y
Answer:
1. 3x = 153770.0
2. 2y = 181436.0
3. 3x + 2y = 335206.0
```

Fine-Tuning Data Construction: The generated outputs are used to construct training examples, where each intermediate step is concatenated with the final correct answer using the separator token `'#|>'`. For instance, from the example above, the following four training data points are created:

1. `"x = 51290.0, y = 90718.0. Please calculate the following: 1. 3x 2. 2y 3. 3x + 2y Answer: #|> 3x + 2y = 335306.0"`
2. `"x = 51290.0, y = 90718.0. Please calculate the following: 1. 3x 2. 2y 3. 3x + 2y Answer: || 1. 3x = 153770.0 #|> 3x + 2y = 335306.0"`
3. `"x = 51290.0, y = 90718.0. Please calculate the following: 1. 3x 2. 2y 3. 3x + 2y Answer: || 1. 3x = 153770.0 || 2. 2y = 181436.0 #|> 3x + 2y = 335306.0"`
4. `"x = 51290.0, y = 90718.0. Please calculate the following: 1. 3x 2. 2y 3. 3x + 2y Answer: || 1. 3x = 153770.0 || 2. 2y = 181436.0 || 3. 3x + 2y = 335206.0 #|> 3x + 2y = 335306.0"`

Each step concatenates the current state of reasoning with the correct final answer. This process enables the supervisor model to learn the relationship between intermediate steps and the correct final outcome.

Using the dataset generated above, we fine-tune a Llama-3-8b model using Low Rank Adaptation (LoRA) [13] as the supervisor model. Finally, the information-gain is computed using the trained model as described in Section C.1.4.

C.2.2 Math Shepherd Results

The Math-Shepherd approach [29] evaluates how well the model generates intermediate results and completes the reasoning process step-by-step. For a given model generation, we iteratively cut off the chain of reasoning after each step and obtain multiple completions using a completer model (in this case, also the Llama-3-8B model).

Consider the following model generation:

```
x = 51290.0, y = 90718.0. Please calculate the following:
1. 3x
2. 2y
3. 3x + 2y
Answer: 1. 3x = 153770.0, 2. 2y = 181436.0, 3. 3x + 2y = 335206.0
```